

Roll No. 20

Total No. of Questions : 9]
(1048)

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**B.C.A. (CBCS) RUSA IInd Semester
Examination**

4027

MATHEMATICS-II

Paper : BCA-0201

Time : 3 Hours]

[Maximum Marks : 70

Note : Attempt five questions in all, selecting one question from each Unit-I to IV. Question No. 9 (Unit-V) is compulsory. All questions are of equal marks.

Unit-I

1. (a) Verify Lagrange's mean value theorem for the function f defined by $f(x) = x^3 - x^2 - 6x$ in the interval $[-1, 4]$.

- (b) Find n th derivative of $e^{ax} \sin (bx + C)$.

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$y = e^{ax} \sin (bx + C)$
 $y_n = e^{ax} \sin (bx + C + n\pi/2)$

Turn Over

2. (a) Discuss the applicability of Cauchy's mean value theorem to $f(x)$ and $g(x)$ in $[a, b]$ where :

$$f(x) = \begin{cases} 2, & a \leq x < b \\ 4, & x = b \end{cases}$$

and $g(x) = x, x \in [a, b]$.

- (b) Find the n th derivative of co-efficient of $x^3 e^x \cos x$.

Unit-II

3. (a) If a and b are any two integers, with $b > 0, a \neq 0$, then there exists unique integers q and r such that $a = bq + r, 0 \leq r < |b|$.

- (b) Solve the linear congruence equation :

$$11x = 2 \pmod{317}$$

4. (a) Find the least member and greatest member of the following set X , if they exist

$$X = \{n \in \mathbb{N} : n^2 + 2n \leq 60\}$$

- (b) For any $n \in \mathbb{I}_+$, prove that the integers $8n + 3$ and $5n + 2$ are relatively prime.

Unit-III

5. (a) Prove that the set $G = \{0, 1, 2, 3, 4\}$ is a finite abelian group of order 5 w.r.t. addition modulo 5.

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$f(x)$

- (b) Show that the set Z of integers, with the ordinary addition operation, is an infinite cyclic group.
6. (a) Prove that the set G_1 of all n -rowed non-singular matrices over a field F is a non-Abelian group w.r.t. the operation of matrix multiplication.
- (b) Prove that $(ab)^{-1} = b^{-1}a^{-1} \forall a, b \in G$, the inverse of the product of two elements of a group G is the product of inverse taken in reverse order.

Unit-IV

7. (a) Prove that a finite ring without zero divisors is a field.
- (b) Prove that in general the set of all numbers of the form $a + (\sqrt{p})b$ where a, b are rational numbers, and p is a prime is a field w.r.t. ordinary addition and multiplication.
8. (a) Let F be a field and suppose $f(x)$ is a polynomial in $F(x)$, then prove that $x - a$ is a divisor of $f(x)$ in $F_1(x)$ iff $f(a) = 0$ in F .
- (b) If R is a ring such that $a^2 = a \forall a \in R$, prove that :
- (i) $a + a = 0 \quad \forall a \in R$
- (ii) R is commutative ring

Unit-V

(Compulsory Question)

9. (a) State Rolle's theorem.
- (b) Find n th derivative of $\frac{1}{ax+b}$ (or)
- (c) Find greatest common divisor of 258, 60 by using Euclidean algorithm.
- (d) If a, b, c and d are integers such that $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then prove that $a + c \equiv b + d \pmod{m}$.
- (e) Prove that every cyclic group is an abelian group.
- (f) Prove that the inverse of each element is unique.
- ~~(g) Show that the set $\mathbb{Z}$ of all integers $\dots, -4, -3, -2, -1, 0, 1, 2, 3, 4, \dots$ is a group w.r.t. the operation of addition of integers.~~
- (h) Define a ring and give two examples.
- (i) If F is a field and $f(x)$ is a polynomial of degree $n \geq 1$ in $F(x)$ then the equation $f(x) = 0$ has at most n roots in F .
- (j) Prove that the set $\mathbb{I}$ of all integers is not a group under addition and multiplication as ring composition.

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$-4 -3 -2 -1 0 1 2 3 4$

$1-2 \quad 1+10 = 11$

$2-11$